

Question 2

a) Price Elasticity Demand (PED) = $\frac{\% \Delta Q}{\% \Delta P}$ — ①

$$\begin{aligned} \% \Delta Q &= \frac{10000}{55000} \times 100 \\ &= \frac{1000}{55} \\ &= 18.18\% \end{aligned}$$

$$\begin{aligned} \% \Delta Q &= \frac{\Delta Q}{Q} \times 100 \\ &\text{initial } Q_1 = 50000 \rightarrow \text{New } Q_2 = 60000 \\ \text{Average } Q &= 55000 \\ &\hookrightarrow = \frac{Q_1 + Q_2}{2} \end{aligned}$$

$$\begin{aligned} \% \Delta P &= \frac{200}{1100} \times 100 \\ &= 18.18\% \end{aligned}$$

$$\begin{aligned} P_1 &= 1200 \rightarrow P_2 = 1000 \\ \text{Average } P &= 1100 \\ &\hookrightarrow = \frac{P_1 + P_2}{2} \end{aligned}$$

②

$$\begin{aligned} PED &= \frac{18.18\%}{18.18\%} \\ &= 1 \end{aligned}$$

$PED = 1$ $\therefore$ unit elastic (unitary)

Total revenue remains unchanged when price changes

b)

$$\text{Cross Elasticity of Demand (XED)} = \frac{\% \Delta Q_{\text{streamFlix}}}{\% \Delta P_{\text{Netflix}}}$$

Streamflix subscribers increase + 18.18 %.

$$XED = \frac{18.18\%}{-5\%}$$

$$= -3.64$$

$XED < 0$ (XED is negative) . therefore suggests the services are complement. = (No relationship)

c)

$$\text{Income Elasticity demand (YED)} = \frac{\% \Delta Q}{\% \Delta Y}$$

$$= \frac{20\%}{12\%}$$

$$= 1.667$$

$YED > 1$ therefore this is a normal luxury good $\therefore$ Demand is highly responsive to income change.

d) Pricing strategy

$PED = 1$, then price change won't increase revenue significantly. therefore this is not affected

Competitive strategy

There is no relation. Two product are not substitute.